

Overall study workflow for cross-species synovial senescence analysis after ACL injury

Mouse discovery of a persistent program → early pig validation → chronic pig extension

Mouse discovery

GSE271903 mouse synovium, paired ACLR vs Contra at 1W and 4W
Paired DE analysis at 1W and 4W
Shared strict DE genes at 1W & 4W: 1,421
Persistent same-directional DE genes: 1,416
Persistent same-directional DE genes n CellAge: 78
Projected CellAge score
Hallmark GSEA: inflammatory response and EMT / ECM remodeling



Persistent program validation in early pig

E-MTAB-6664 early pig synovium, Control vs ACLT at 1W and 4W
78 human genes projected to pig ortholog space
75 detected in pig; 47 direction-consistent
Persistent senescence-associated signature-score validation
Hallmark GSEA: inflammatory response and EMT / ECM remodeling
Early-defined strict core ortholog genes: 24



Pig chronic extension

GSE228848 chronic pig synovium, Control_52W vs ACLT_alone_52W
Persistent senescence-associated signature-score validation
Hallmark GSEA: inflammatory response and EMT / ECM remodeling
Retention audit of early-defined core genes performed
No chronic-specific signature redefinition